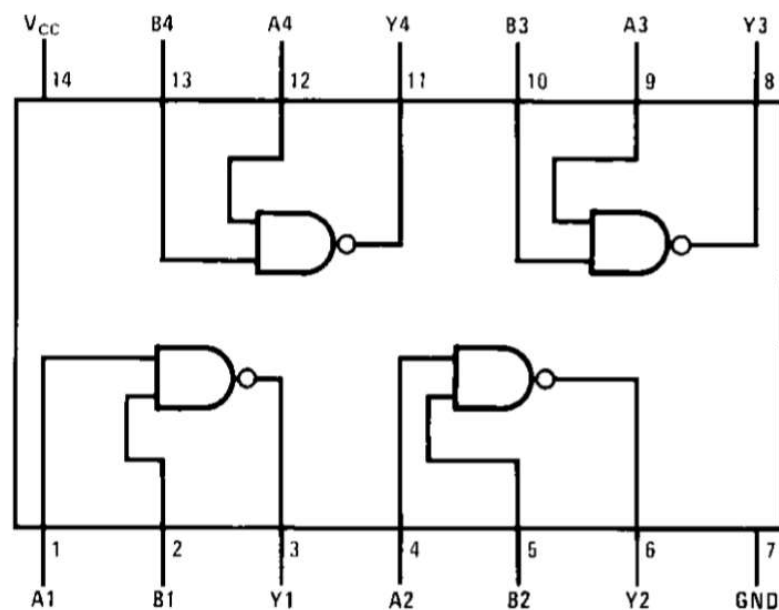


组合逻辑实验所用的集成电路

① 74LS00——四个2输入的与非门

Connection Diagram



Function Table

$$Y = \overline{AB}$$

Inputs		Output
A	B	Y
L	L	H
L	H	H
H	L	H
H	H	L

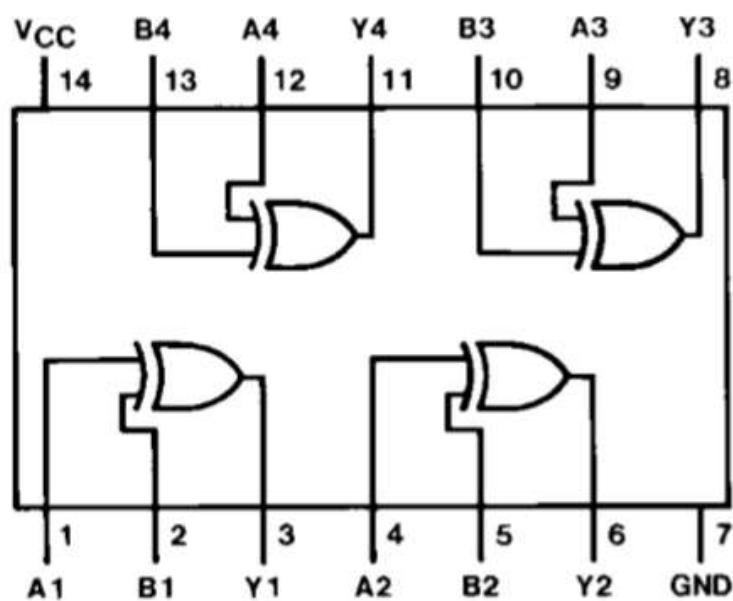
H = HIGH Logic Level

L = LOW Logic Level

组合逻辑实验所用的集成电路

② 74LS86——四个2输入的异或门

Connection Diagram



Function Table

$$Y = A \oplus B = \bar{A}B + A\bar{B}$$

Inputs		Output
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	L

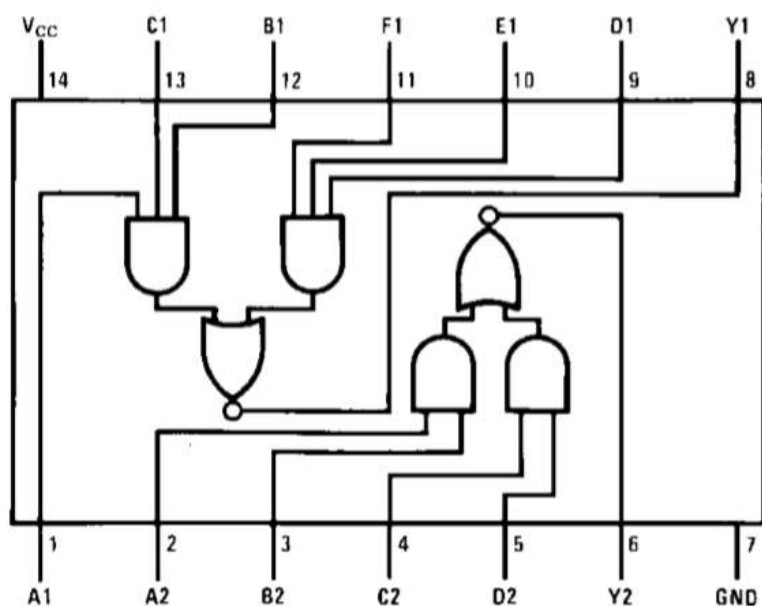
H = HIGH Logic Level

L = LOW Logic Level

组合逻辑实验所用的集成电路

③ 74LS51——两个与或非门

Connection Diagram



Function Table

$$Y1 = \overline{(A1)(B1)(C1) + (D1)(E1)(F1)}$$

Inputs						Output
A1	B1	C1	D1	E1	F1	Y1
H	H	H	X	X	X	L
X	X	X	H	H	H	L
Other Combinations						H

$$Y2 = \overline{((A2)(B2) + (C2)(D2))}$$

Inputs				Output
A2	B2	C2	D2	Y2
H	H	X	X	L
X	X	H	H	L
Other combinations				H

H = HIGH Logic Level

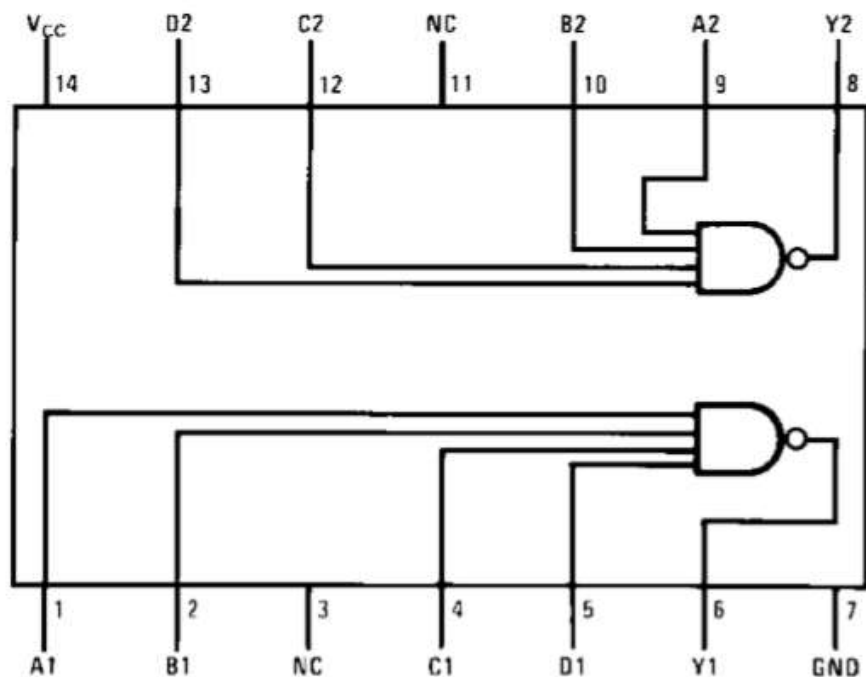
L = LOW Logic Level

X = Either LOW or HIGH Logic Level

组合逻辑实验所用的集成电路

④ 74LS20——两个4输入的与非门

Connection Diagram



Function Table

$$Y = \overline{ABCD}$$

Inputs				Output
A	B	C	D	Y
X	X	X	L	H
X	X	L	X	H
X	L	X	X	H
L	X	X	X	H
H	H	H	H	L

H = HIGH Logic Level

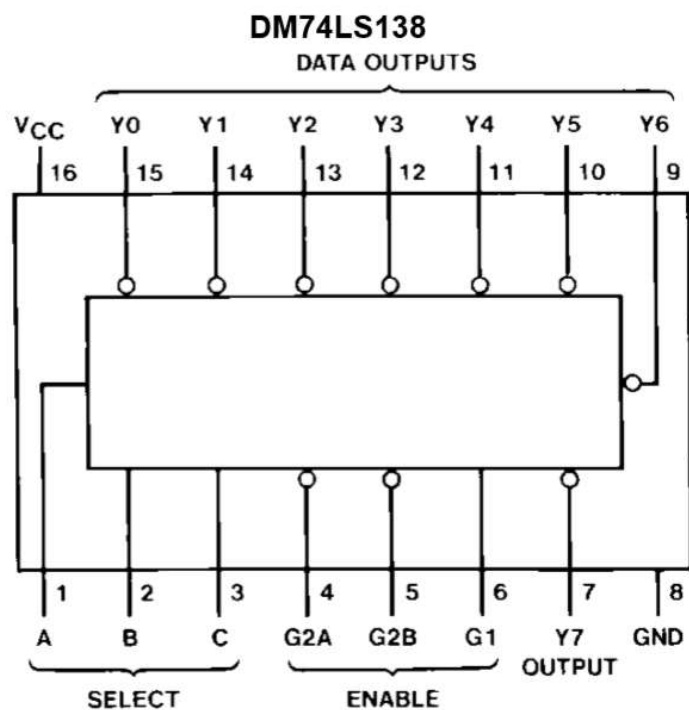
L = LOW Logic Level

X = Either LOW or HIGH Logic Level

译码器实验所用的集成电路

74LS138——译码器/多路分配器

Connection Diagrams



Function Tables

DM74LS138

Inputs			Outputs									
Enable		Select										
G1	G2 (Note 1)	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	H	H	H	H	H	H	H	H
H	L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	H	H	L	H	H	H	H	H	H
H	L	L	H	L	H	H	L	H	H	H	H	H
H	L	L	H	H	H	H	H	L	H	H	H	H
H	L	H	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	H	L	H	H
H	L	H	H	L	H	H	H	H	H	H	L	H
H	L	H	H	H	H	H	H	H	H	H	H	L

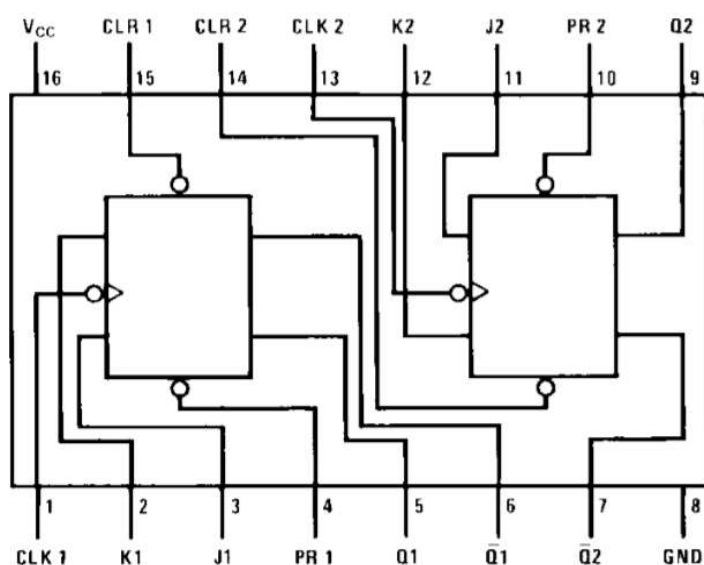
H = HIGH Level
L = LOW Level
X = Don't Care

Note 1: G2 = G2A + G2B

触发器实验所用的集成电路

① 74LS112——Dual J-K Negative-edge-triggered Flip-Flops(with Preset and Clear)

Connection Diagram



Function Table

Inputs					Outputs	
PR	CLR	CLK	J	K	Q	$\bar{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H (Note 1)	H (Note 1)
H	H	↓	L	L	Q_0	$\bar{Q}_0$
H	H	↓	H	L	H	L
H	H	↓	L	H	L	H
H	H	↓	H	H	Toggle	
H	H	H	X	X	Q_0	$\bar{Q}_0$

保持
置1
置0
反转

注：针对输出Q而言

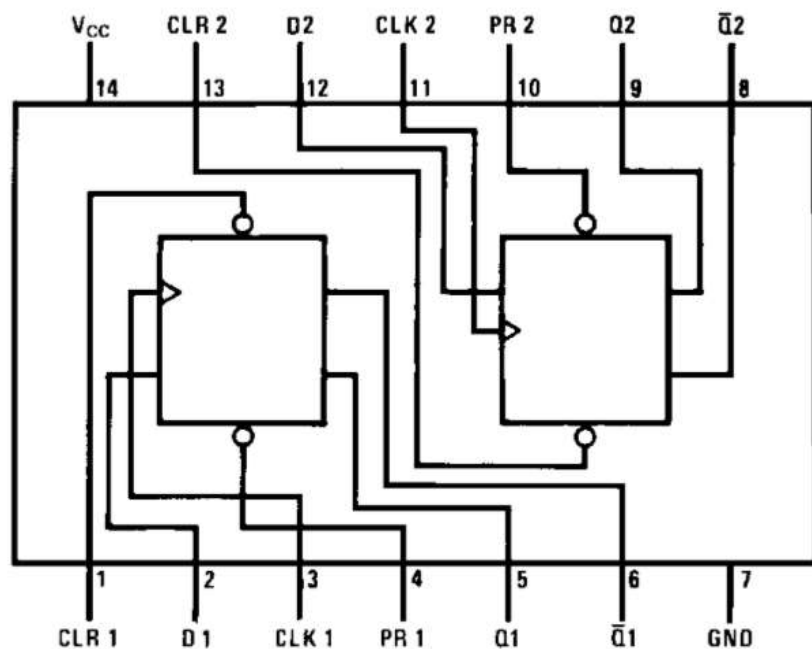
H = HIGH Logic Level
L = LOW Logic Level
X = Either LOW or HIGH Logic Level
↓ = Negative Going Edge of Pulse
 Q_0 = The output logic level before the indicated input conditions were established.
Toggle = Each output changes to the complement of its previous level on each falling edge of the clock pulse.

Note 1: This configuration is nonstable; that is, it will not persist when preset and/or clear inputs return to their inactive (HIGH) level.

触发器实验所用的集成电路

② 74LS74——Dual Positive-Edge-Triggered D Flip-Flops with Preset, Clear and Complementary Outputs

Connection Diagram



Function Table

Inputs				Outputs	
PR	CLR	CLK	D	Q	$\bar{Q}$
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H (Note 1)	H (Note 1)
H	H	$\uparrow$	H	H	L
H	H	$\uparrow$	L	L	H
H	H	L	X	Q_0	$\bar{Q}_0$

H = HIGH Logic Level

X = Either LOW or HIGH Logic Level

L = LOW Logic Level

$\uparrow$ = Positive-going Transition

Q_0 = The output logic level of Q before the indicated input conditions were established.

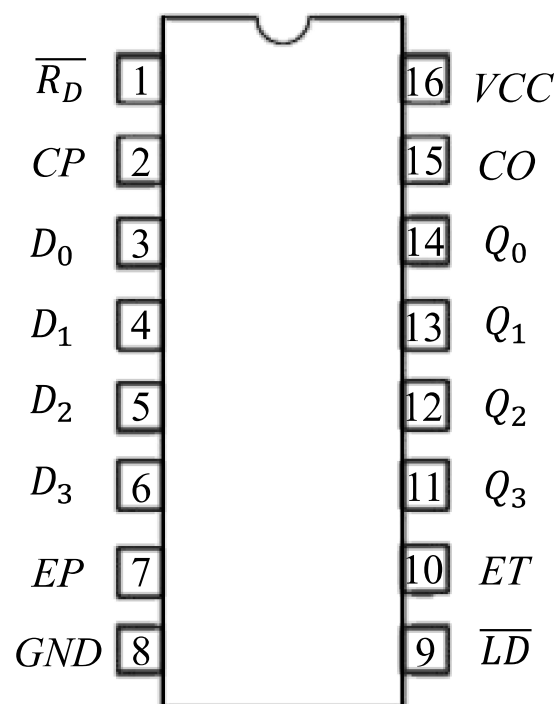
Note 1: This configuration is nonstable; that is, it will not persist when either the preset and/or clear inputs return to their inactive (HIGH) level.

置1
置0

注：针对输出Q而言

计数器实验所用的集成电路

74LS160——4 bit binary counters/BCD decade counters



输入									输出				工作状态
2	1	9	7	10	3	4	5	6	14	13	12	11	
CP	$\overline{R_D}$	$\overline{LD}$	EP	ET	D ₀	D ₁	D ₂	D ₃	Q ₀	Q ₁	Q ₂	Q ₃	
×	L	×	×	×	×	×	×	×	L	L	L	L	异步清零
↑	H	L	×	×	a	b	c	d	a	b	c	d	预置数
↑	H	H	H	H	×	×	×	×	二进制加法计数				计数
×	H	H	L	×	×	×	×	×	保持				保持
×	H	H	×	L	×	×	×	×	保持(但CO=0)				保持

CP为脉冲输入端, $\overline{R_D}$ 为清零端, $\overline{LD}$ 为置数端, EP、ET为保持端
 D₀, D₁, D₂, D₃为输入端, Q₀, Q₁, Q₂, Q₃为输出端
 CO为十进制进位端